

How Accurately Can One Measure a Statistical Quantity Like Polarization-Mode Dispersion?

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Abstract—The differential group delay (DGD) due to varying local birefringence and random polarization mode coupling is a statistical quantity. Its value at any wavelength fluctuates in time. At any given moment it is wavelength sensitive. Hence, polarization-mode dispersion can only be defined as a mean or rms value of the DGD. All measurements of polarization-mode dispersion involve thus an averaging over a finite sample set. Consequently, all measurements are intrinsically of limited accuracy. We present analytical formulae and numerical data characterizing this intrinsic limitation common to all measurement techniques.

POLARIZATION-MODE dispersion (PMD) limits the bandwidth of digital and the linearity of analog optical transmissions. Different measurement techniques of PMD are presently available and under further investigations [1]–[4]. All these techniques involve an averaging procedure. Indeed, this is necessary, since the differential group delay (DGD) is a statistical quantity and PMD is defined either as the mean DGD or its rms value. However, the statistical character of the DGD and the fact that any measurement involves only a finite sample set imply that there is an intrinsic limitation to the accuracy of PMD measurements. This limitation is independent of any measurement setup implementation, but is unavoidable, similarly to the well-known example of the measurement of the probability of coin tossing with only a finite number of trials. The main difficulty in evaluating this limitation in the case of PMD is that, contrary to the coin-tossing example, the DGD at nearby wavelengths are not independent.

Before presenting our results, let us emphasize the importance of understanding such intrinsic limitations. Indeed, nowadays most parameters of optical fibers (like attenuation, mode field diameter, geometrical parameters, etc.) are currently measured with high accuracy, while PMD resists such high accuracy [5]. But we shall see that the present-day accuracy of available instruments is already close to the theoretical limit. Hence, no major improvement is to be expected without significantly expanding the sample set, i.e., expanding the wavelength range of the measurements or the measurement time, or varying the environmental conditions during the measurement.

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In order to present our results, let us introduce some notations. Let $\vec{\Omega}(\omega)$ denote the principal polarization state vector [6]. By convention, its norm equals the DGD: $\delta\tau(\omega) = \sqrt{\Omega_x^2(\omega) + \Omega_y^2(\omega) + \Omega_z^2(\omega)}$. In practice, one is interested in statistical ensembles (i.e., similar fibers) or in time ensembles. However, wavelength ensembles are much easier to produce. Furthermore, it is generally admitted that these different ensembles are statistically equivalent and there are good reasons [7] for doing so. Consequently, all major PMD measurement techniques are based on a wavelength ensemble [9]. Note that large wavelength ranges correspond to long time fluctuations. Accordingly, we define the PMD delay $\Delta\tau$ as the rms value of the $\delta\tau(\omega)$:

$$\Delta\tau(\omega_1, \omega_2)^2 = \frac{1}{\omega_2 - \omega_1} \int_{\omega_1}^{\omega_2} \delta\tau(\omega)^2 d\omega. \quad (1)$$

In practice, the optical frequency range is limited by the laser tunability or to the spectral width of the LED. Another limitation is due to the slight wavelength dependence of the modal birefringence and the polarization-mode coupling length. However, in order to define a PMD value valid for arbitrary long-time fluctuations, we shall assume in our model that the local modal birefringence and the mean polarization-mode coupling length are wavelength independent. This amounts to assume that the modal field does not change over the considered wavelength range (i.e., all wavelengths see the same refractive index structure), hence, the wavelength dependence of $\delta\tau(\omega)$ is only due to interferences between the different optical trajectories in the fiber [4]. With such an assumption, the theoretical mean PMD delay $\overline{\Delta\tau}$ can be defined as

$$\overline{\Delta\tau} = \lim_{\omega_2 \rightarrow \infty} \Delta\tau(0, \omega_2). \quad (2)$$

Defining PMD this way has the great advantage that the probability space is well defined and the statistical ensemble of interest consists in all optical frequency intervals of length $\Delta\omega$.

Henceforth, we study the variance of $\Delta\tau(\omega_1, \omega_2)^2$ when $\Delta\omega \equiv \omega_2 - \omega_1$ is finite (recall that the variance is defined as $\text{Var}(X) = \text{mean}(X^2) - \text{mean}(X)^2$).

Let ω_j , $j = 1 \dots N$ denote the optical frequencies at which the DGD $\delta\tau(\omega_j)$ are measured. Hence, this finite sample set corresponds to a measured value of PMD $\Delta\tau_{\text{meas}}$:

$$\begin{aligned} \Delta\tau_{\text{meas}}^2 &= \frac{1}{N} \sum_j \delta\tau(\omega_j)^2 \\ &= \frac{1}{N} \sum_j (\Omega_x(\omega_j)^2 + \Omega_y(\omega_j)^2 + \Omega_z(\omega_j)^2) \quad (3) \end{aligned}$$

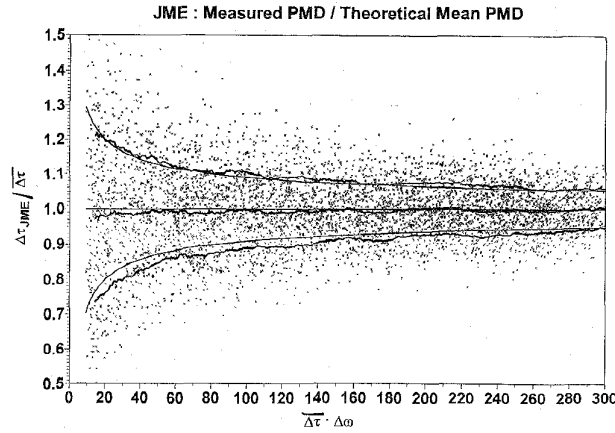


Fig. 1. Illustration of our main result (6) on 5000 numerical simulations for the case of the Jones Matrix Eigenanalysis (JME) measurement method simulating a tunable laser over 90 nm around 1300 nm (i.e., $\Delta\omega = 100 \frac{1}{\text{ps}}$). Each point represents one run of the software, simulating a concatenation of 200 polarization maintaining fibers of random orientation and birefringence. Hence, all the polarization-mode coupling takes place at the junction between the trunks. The distribution of each component of the local modal birefringence vectors was assumed Gaussian. The theoretical mean PMD value $\Delta\tau$ was computed using (1) over a frequency range $50,000 \frac{1}{\text{ps}}$. $\Delta\tau_{\text{JME}}$ is computed according to published procedures [2], using the definition (1). The thick lines are the computed mean values (over intervals of length 20 on the abscissa) and the mean plus and minus one standard deviation. The thinner lines are the theoretical curves (6) with $\alpha = 0.9$.

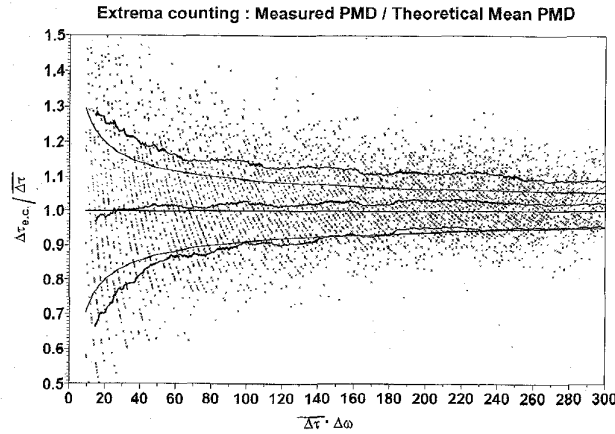


Fig. 2. Same as Fig. 1, but for the extrema counting measurement method: $\Delta\tau_{e.c.} = \sqrt{\frac{3\pi}{8}} 0.824\pi \frac{N_e}{\Delta\omega}$ where N_e is the number of extrema in the interval $\Delta\omega = 100 \frac{1}{\text{ps}}$.

where each component of the principal state vector $\vec{\Omega}$ is a normal random variable [8] of zero mean and variance $\Delta\tau^2/3$. Hence, $\Delta\tau_{\text{meas}}^2$ is the sum of three normal random variables of mean $\Delta\tau^2/3$. Recall that for any normal random variable X , $\text{Var}(X^2) = 2\text{Var}(X)^2$ and that the variance of the sum of independent random variables is simply the sum of the variances. Hence, if the $\Omega_k(\omega_j)$ would be independent, the variance of $\Delta\tau_{\text{meas}}^2$ would be equal to $\frac{2\Delta\tau^4}{3N}$, but, clearly, they are not independent. The number of data that are effectively independent can, however, be estimated as follows.

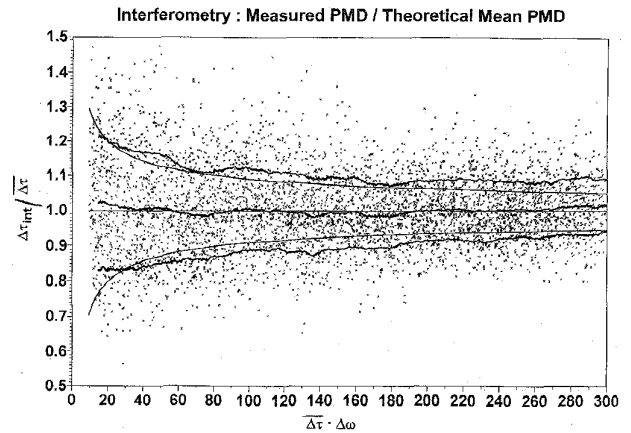


Fig. 3. Same as Fig. 1, but for the interferometric measurement method (deducing $\Delta\tau_{\text{int}}$ from the second moment of the interferogram with Heffner's [10] factor $\sqrt{\frac{3}{4}}$). The source spectrum was assumed Gaussian with a FWHM of $\Delta\omega = 80 \frac{1}{\text{ps}}$.

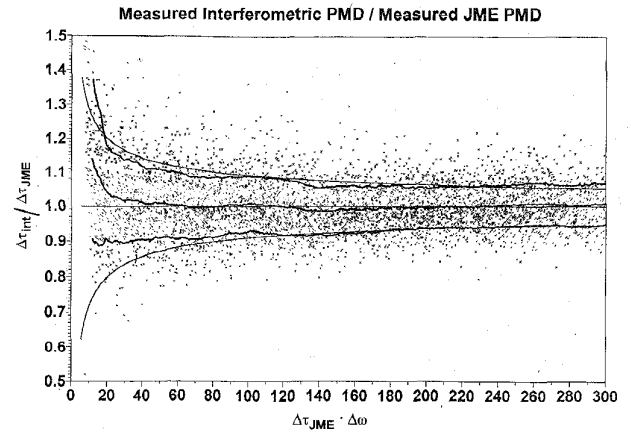


Fig. 4. Comparison of the JME and interferometric measurements. Note that the standard deviation of this ratio is identical to that of the previous graphs (the thin lines are the same on all graphs). This illustrates that the uncertainty is similar for all measurement methods.

The functions $\Omega_k(\omega)$, $k = x, y, z$ have irregular oscillations, hence, using the Nyquist theorem two points per cycle can be considered as independent. Similarly to the well-known relation between the extrema of the intensity passing through a fixed analyzer and the PMD [3], our numerical simulation showed that the number N_e of extrema of $\Omega_k(\omega)$ is related to the PMD as

$$\Delta\tau \approx \sqrt{\frac{3\pi}{8}} \pi \frac{N_e}{\Delta\omega}. \quad (4)$$

Hence, we arrive at our main result

$$\text{Var}(\Delta\tau_{\text{meas}}^2) \approx \sqrt{\frac{3\pi}{8}} \frac{2\pi}{3} \frac{\Delta\tau^3}{\Delta\omega} \approx 2.3 \frac{\Delta\tau^3}{\Delta\omega}. \quad (5)$$

Consequently, developing $\Delta\tau_{\text{meas}}^2 \approx \Delta\tau^2 \pm \sqrt{\text{Var}(\Delta\tau_{\text{meas}}^2)}$ to first order, a measurement of the DGD over a wavelength range

$\Delta\omega$ provides an approximate value $\Delta\tau_{\text{meas}}$ of the theoretical mean PMD delay $\overline{\Delta\tau}$:

$$\Delta\tau_{\text{meas}} \approx \overline{\Delta\tau} \left(1 \pm \frac{\alpha}{\sqrt{\overline{\Delta\tau} \Delta\omega}} \right) \quad (6)$$

where $\alpha = \frac{\sqrt{2.3}}{2} \approx 0.76$. A more elaborate derivation of relation (6) (see the Appendix) confirms this relation, but with $\alpha \approx 0.9$.

For example, if $\Delta\omega = 100 \frac{1}{\text{ps}}$ (corresponding in wavelengths to $\delta\lambda \approx 130$ nm around 1550 nm or to $\delta\lambda \approx 90$ nm around 1300 nm), then for a PMD of 1 ps, the measurement uncertainty due to the statistical nature of the DGD is of about 9%! For a PMD of 0.1 ps, the uncertainty is even as large as 28%. Note that for a Maxwell distribution, one standard deviation equals about 40% of the rms value. Hence, for a PMD of 0.1 ps, any PMD measurement limited to about 100 nm provides almost no more information than a measurement of the DGD at a single wavelength, as could be expected from the fact that the DGD bandwidth [8] for such small PMD is close to 100 nm.

It should be stressed that the above limitation (6) is not specific to one measurement scheme, but is intrinsic to all measurement techniques that average the DGD over a wavelength range. All four methods presently under investigation at the major standardization bodies (ITU, TIA, IEC, etc.) fall in this large class and therefore are restricted by the limitation (6). For example, Figs. 1–3 illustrate our result on simulations. Fig. 1 presents the case of the Jones Matrix Eigenanalysis (JME) measurement method simulating a tunable laser over 90 nm around 1300 nm. Each point represents one run of the simulation software. The thick lines are the mean values and the mean plus and minus one standard deviation. The thinner lines are the theoretical curves with $\alpha = 0.9$. Figs. 2 and 3 are analogous, but for the extrema counting and interferometric measurement schemes, respectively. Fig. 4 shows the ratio of simulated measurement results as obtained with the JME and interferometric methods.

In conclusion, a general formula (6) of the accuracy limitation to PMD measurement due to the statistical nature of the DGD has been derived. This was possible thanks to a clear definition of the probability space. The formula agrees well with numerical simulations. Hence, the results are intrinsic, independent of any technical limits and essentially independent of the measurement method. Present-day commercial instruments are already close to the predicted theoretical accuracy limitation.

APPENDIX

In this appendix, we briefly present a more rigorous derivation of our result based on an invariance law of the correlation

function. Let

$$\begin{aligned} \text{Var}(\Delta\tau(\omega_1, \omega_2)^2) &= E((\Delta\tau(\omega_1, \omega_2)^2 - \overline{\Delta\tau}^2)^2) \\ &= \frac{1}{(\omega_2 - \omega_1)^2} \int_{\omega_1}^{\omega_2} \int_{\omega_1}^{\omega_2} C_{\overline{\Delta\tau}}(\omega - \omega') d\omega d\omega' \end{aligned} \quad (7)$$

where E denotes the expectation value and the correlation function

$$C_{\overline{\Delta\tau}}(\omega - \omega') = E((\delta\tau(\omega)^2 - \overline{\Delta\tau}^2)(\delta\tau(\omega')^2 - \overline{\Delta\tau}^2)). \quad (8)$$

From the dynamical equation for PMD [4, (9)], one deduces the invariance law: $C_{\overline{\Delta\tau}}(\omega) = \overline{\Delta\tau}^4 C_1(\omega \overline{\Delta\tau})$, where C_1 is defined by (8) with $\overline{\Delta\tau} = 1$. Hence, one obtains

$$\text{Var}(\Delta\tau(\omega_1, \omega_2)^2) = 4\alpha^2 \frac{\overline{\Delta\tau}^3}{\omega_2 - \omega_1} - \beta \frac{\overline{\Delta\tau}^2}{(\omega_2 - \omega_1)^2} \quad (9)$$

where $4\alpha^2 = \int_0^{2\omega \overline{\Delta\tau}} C_1(x) dx$ and $\beta = \int_0^{2\omega \overline{\Delta\tau}} x C_1(x) dx$. The second term is only a small correction to the first one. Note that whenever $2\omega \overline{\Delta\tau}$ is large, α and β are constant. Hence, this more rigorous derivation recovers our main result (5).

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